



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering.
Academic Year: 2023- 2024(Odd/Even Semester)

Degree, Semester & Branch: B.E,III & EEE

Course Code & Title: CS3353 & C programming and Data Structure

Name of the Faculty member (s): A. Sofia Thanga Mary, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Hashing Techniques**
- **Course Outcome: CO4**
- **Topic Learning Outcome: TLO11**
 - **Activity Chosen: Reflection**
 - **Justification:**
 - Hashing is an important concept in data structures. Different types of hashing collision resolving techniques were taught in previous session.
 - Reflection is a process where teachers can identify their own teaching process and analysing how their teaching (content delivery) might be improved for the better understanding of the students. The prime objective of Reflection is to make the students write their doubts in a paper on topics which they are not clear, so as the teacher can discuss and clarify the concepts that are not understood by the students.
- **Time Allotted for the Activity: 20 Minutes**
- **Details of the Implementation:**

After I completed the hashing, in order to address the doubts and to know the level of understanding of the students, they are insisted to honestly describe the concepts that they didn't understand well. I asked the students to write their doubts in hashing techniques.

The students wrote their doubts and topics which they are not clear. The papers are collected by me and I allotted 10 minutes time in the next hours and discussed all the doubts in next class session and make them understand. I also gave a few simple tasks in classroom to gain more understanding in the topics. The students felt they understood the concepts better.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1	PSO2
	3	3	2	2	1	1	2	1	2	2

- **PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PSO10	PSO1
	3	3	2	2	2	2
Justification for correlation	Requires basic engineering knowledge to implement hash table	Able to Identify engineering problems with hashing resolution techniques	Able to design solutions for complex engineering problems using hashing resolution technique	Able to solve collision in hashing individually or in team and implement hash table	Able to explain the way to resolve collision in hashing technique.	Able to apply hash data structures to solve various application in Information & communication Industry

• Images / Screenshot of the practice:

Reflection Activity.
Aarthi Sri

not understand
1. What are the steps in double hashing?

Understand.
1. Linear Probing
2. rehashing
3. separate chaining.



Shubas

1. I haven't understood separate chaining concept

Understand
1. Linear Probing
2. Double Hashing
3. Rehashing.



Reflective Critique:**❖ Feedback of practice from students and other stakeholders:**

- Students felt the activity conducted gave them a chance to represent their common doubts among students and could get additional clarification from me.

❖ Benefit of the practice:

- Through this activity the students can recollect the topics on hashing techniques. The students are made to understand the concepts in which they are not clear and ensured that they understood the topics thoroughly.

❖ Challenges faced in implementation:

- Students were not able to frame or express their doubts in English sentence properly and some questions were not clear to understand.
- Slow learners were not interested to participate in this innovative teaching and submitted the empty paper.
- Some of the students did not even ready to write their doubts
- Students started to discuss and shout with their neighbours during the activity.

References:

- https://apiar.org.au/wpcontent/uploads/2017/02/13_APJCECT_Feb_BRR798_EDU-126-131.pdf
- <https://www.skillsyouneed.com/ps/reflective-practice.html>
- <https://www.cambridge-community.org.uk/professionaldevelopment/gswrp/index.html>

Signature of Faculty Member

HOD